of the twentieth century, led a brutal personal attack on Shechtman. In a conference attended by hundreds of scientists, Pauling proclaimed, 'Danny Shechtman is talking nonsense, there are no quasicrystals, just quasi-scientists.'

But Shechtman was not imprisoned or killed. He got a place in another lab. The evidence he presented turned out to be more convincing than the existing chemistry textbooks and the views of Linus Pauling. Several colleagues repeated Shechtman's experiments and replicated his findings. A mere ten years after Shechtman saw the quasicrystals through his microscope, the International Union of Crystallography – the leading scientific association in the field – altered its definition of what a crystal is. Chemistry textbooks were changed accordingly, and an entire new scientific field emerged – the study of quasicrystals. In 2011, Shechtman was awarded the Nobel Prize in Chemistry for his discovery.[113] The Nobel Committee said that 'his discovery was extremely controversial [but] eventually forced scientists to reconsider their conception of the very nature of matter'.[114]

Shechtman's story is hardly exceptional. The annals of science are full of similar cases. Before the theory of relativity and quantum mechanics became the cornerstones of twentieth-century physics, they initially provoked bitter controversies, including personal assaults by the old guard on the proponents of the new theories. Similarly, when Georg Cantor developed in the late nineteenth century his theory of infinite numbers, which became the basis for much of twentieth-century mathematics, he was personally attacked by some of the leading mathematicians of his day, like Henri Poincaré and Leopold Kronecker. Populists are right to think that scientists suffer from the same human biases as everyone else. However, thanks to institutional self-correcting mechanisms these biases can be overcome. If enough empirical evidence is provided, it often takes just a few decades for an unorthodox theory to upend established wisdom and become the new consensus.

As we shall see in the next chapter, there were times and places where scientific self-correcting mechanisms ceased functioning and academic dissent *could* lead to physical torture, imprisonment and death. In the Soviet Union, for example, questioning official dogma on any matter – economics, genetics or history – could lead not only to dismissal but even to a couple of years in the gulag or an executioner's bullet.[115] A famous case involved the